WORKED SOLUTION

STEP II 2014, Question 9

Mechanics • Rigid Body Equilibrium

STEP II 2014, Question 9

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/14-s2-q9

The key idea

There are three unknown forces (the tension, the normal reaction and the friction at A) and three equations: resolve horizontally, resolve vertically, and take moments. Taking moments about the corner A is the clever choice, because both contact forces at the wall act there and drop out, leaving a single equation linking the tension to the weight. Once that is in hand, the direction of friction is just a question of whether the string alone holds up more or less than the weight.

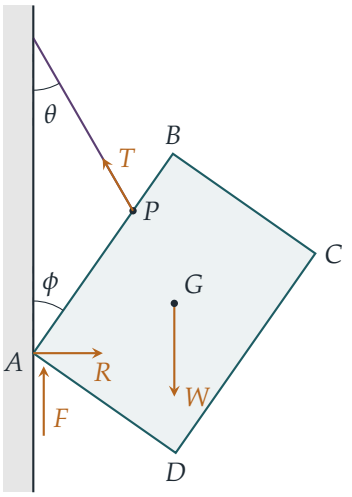
Solution

We model the lamina as a uniform rigid body of weight W with its centre of mass G at the centre of the rectangle, and the string as light and inextensible, so the tension T is the same along it. At A the wall exerts a normal reaction R (horizontal, away from the wall) and a friction force F (vertical). We take F to be positive upwards, so a negative F means the friction actually acts downwards.

Set up axes with A at the origin, x horizontal away from the wall and y vertically up. The side AB makes angle ϕ with the upward vertical, so the unit vector along AB is $(\sin\phi, \cos\phi)$. The side AD is perpendicular to it and points away from the wall, so its unit vector is $(\cos\phi, -\sin\phi)$. Hence

$$G = b(\sin\phi, \cos\phi) + a(\cos\phi, -\sin\phi), \qquad P = d(\sin\phi, \cos\phi),$$

where P is the point where the string is attached. The string runs from P up to the wall at angle θ to the downward vertical, so the tension on the lamina is $T(-\sin\theta, \cos\theta)$.



Part (i)

Resolving horizontally and vertically,

$$R = T\sin\theta,$$
$$F + T\cos\theta = W.$$

For moments about A, R and F contribute nothing. The anticlockwise moment of a force (f_x, f_y) at (x, y) is $xf_y - yf_x$. For the tension at P this is

$$d\sin\phi \cdot T\cos\theta - d\cos\phi \cdot (-T\sin\theta) = Td\sin(\theta+\phi),$$

and for the weight $(0, -W)$ at G it is $-W(a\cos\phi + b\sin\phi)$. So

$$Td\sin(\theta+\phi) = W(a\cos\phi + b\sin\phi). \tag{1}$$

The quantity $a\cos\phi + b\sin\phi$ is just the horizontal distance of G from the wall, which is a quick way to see the weight's moment.

In limiting equilibrium with friction upwards, $F = \mu R = \mu T\sin\theta$, so the vertical equation becomes

$$T(\cos\theta + \mu\sin\theta) = W.$$

Substituting W into (1) and cancelling T (which is positive, as the string is taut),

$$d\sin(\theta+\phi) = (\cos\theta + \mu\sin\theta)(a\cos\phi + b\sin\phi),$$

which is $(*)$.

Part (ii)

If the friction is limiting and acts downwards, then $F = -\mu R = -\mu T\sin\theta$ and the vertical equation is $T(\cos\theta - \mu\sin\theta) = W$. Nothing else changes, so $(*)$ becomes

$$d\sin(\theta+\phi) = (\cos\theta - \mu\sin\theta)(a\cos\phi + b\sin\phi);$$

that is, μ is replaced by $-\mu$.

Part (iii)

Now we don't assume the friction is limiting. Equation (1) holds in any equilibrium and gives

$$T = \frac{W(a\cos\phi + b\sin\phi)}{d\sin(\theta+\phi)},$$

and $\sin(\theta+\phi) > 0$ because θ and ϕ are both acute. The vertical equation then gives

$$F = W - T\cos\theta = W\left(1 - \frac{\cos\theta\,(a\cos\phi + b\sin\phi)}{d\sin(\theta+\phi)}\right).$$

? **Ask yourself** How do we turn "$F > 0$" into a condition involving only tangents?

The friction acts upwards exactly when $F > 0$, that is, when

$$d\sin(\theta+\phi) > \cos\theta\,(a\cos\phi + b\sin\phi),$$

since $d\sin(\theta+\phi) > 0$. Expand $\sin(\theta+\phi) = \sin\theta\cos\phi + \cos\theta\sin\phi$ and divide both sides by $\cos\theta\cos\phi$, which is positive because both angles are acute, so the inequality keeps its direction:

$$d(\tan\theta + \tan\phi) > a + b\tan\phi.$$

As $\tan\theta + \tan\phi > 0$, the necessary and sufficient condition is

$$d > \frac{a + b\tan\phi}{\tan\theta + \tan\phi}.$$

(When d equals this value there is no friction at all, and when d is smaller the friction acts downwards.)

Physically, the string is attached to the side AB, so $d \le 2b$. If the condition holds, then

$$\frac{a + b\tan\phi}{\tan\theta + \tan\phi} < d \le 2b,$$

so $a + b\tan\phi < 2b\tan\theta + 2b\tan\phi$, that is, $a < b(2\tan\theta + \tan\phi)$. Taking the contrapositive: if $b(2\tan\theta + \tan\phi) < a$, then no allowed position of the attachment point makes the friction act upwards, so the condition cannot be satisfied. This completes the proof.

Checks

- With $\mu = 0$ the wall is smooth, and $(*)$ and its part (ii) version coincide, as they should. They both reduce to the borderline case $d\sin(\theta+\phi) = \cos\theta\,(a\cos\phi + b\sin\phi)$ of part (iii).
- Lamina with AB vertical ($\phi \to 0$): the borderline becomes $d\tan\theta = a$. This says the string's line of action passes through the vertical line through G, at horizontal distance a from the wall. Then the string and weight can balance alone, with the wall only pushing horizontally.

- Numerically, with $a = 1$, $b = 1.5$, $\theta = 30^\circ$, $\phi = 40^\circ$: the critical value is $d \approx (1 + 1.259)/(0.577 + 0.839) \approx 1.595 < 3 = 2b$. Solving the three equilibrium equations directly with moments about G instead of A gives $F > 0$ just above this value and $F < 0$ just below it. A computer check of thousands of random configurations confirmed $(*)$, its part (ii) form and the condition in (iii).

Common slips

- Taking moments about P or G: it works, but brings R and F into the moment equation and makes the algebra much messier.
- Getting the position of G wrong: D lies on the side away from the wall, so AD points along $(\cos\phi, -\sin\phi)$ and G is at horizontal distance $a\cos\phi + b\sin\phi$ from the wall.
- Forgetting the constraint $d \leq 2b$ in the last part; it is the only place the length of AB enters.

Beyond STEP

This is a small statically determinate problem: three unknown forces and three independent equations of equilibrium for a rigid body in a plane. Engineers check this counting before anything else; with more supports the forces can't be found from statics alone, and you need to know how the body deforms. The trick of taking moments about the point where the unknowns act is the planar version of choosing a clever origin for torques. For more, see Kleppner and Kolenkow, *An Introduction to Mechanics*.

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